manages event registrations

Problem Statement:

A college currently manages **event registrations** through Google Forms and email confirmations. Students face issues such as duplicate entries, no real-time status updates, and last-minute communication gaps.

Problem Summary

Event organizers currently rely on Google Forms and email confirmations for event registrations, leading to several operational challenges:

- **Duplicate entries:** Participants accidentally register multiple times due to lack of system validation
- **No real-time status updates:** Attendees cannot track their registration status or receive instant confirmation
- **Last-minute communication gaps:** Important updates about event changes, venue modifications, or cancellations don't reach attendees promptly
- **Manual data management:** Event organizers must manually review and deduplicate entries, wasting valuable time

- **Limited capacity management:** No automated system to handle waitlists or notify attendees when spots become available

Requirement Candidates for Improved Event Registration System

1. **Automated Registration Validation:** The system shall prevent duplicate registrations by checking participant email addresses and ID numbers before allowing form submission, providing immediate feedback if a registration already exists.
2. **Real-Time Status Dashboard:** Attendees shall have access to a personal dashboard showing their registration status (confirmed, waitlisted, or cancelled), event details, and any updates, with instant notifications pushed via email and mobile app.
3. **Instant Communication Channel:** The system shall include a multi-channel notification system (email, SMS, push notifications) that allows organizers to send urgent updates to all registered participants within minutes, with read receipts and delivery confirmations.
4. **Smart Capacity Management:** The system shall automatically manage event capacity limits, create waitlists when events are full, and automatically promote waitlisted attendees when spots become available, sending immediate notifications to affected participants.
5. **Integrated Check-In System:** The system shall provide QR code-based check-in functionality for event day, allowing organizers to quickly verify attendance, track no-shows, and generate real-time attendance reports for event records.

📊 Interactive System Visualization

💡 **Click on the toggles below to explore each component of the improved registration system**

▼ 🔐 User Registration Flow

```
graph TD
    A["Attendee Access Portal"] --> B{"Already Registered?"};
    B -- Yes --> C["Show Registration Status"];
    B -- No --> D["Enter Details"];
    D --> E{"Duplicate Check"};
    E -- Found --> F["Display Existing Registration"];
    E -- Not Found --> G{"Event Full?"};
    G -- No --> H["Confirm Registration"];
    G -- Yes --> I["Add to Waitlist"];
    H --> J["Send Confirmation"];
    I --> K["Send Waitlist Notification"];
```

Key Features:

- Real-time duplicate detection
- Instant email confirmation
- Automatic waitlist management

▼ 📱 Notification System

```
graph LR
    A["Event Update"] --> B["Notification Engine"];
    B --> C["Email"];
    B --> D["SMS"];
    B --> E["Push Notification"];
    C --> F["Delivery Tracking"];
    D --> F;
    E --> F;
    F --> G["Read Receipts"];
```

Communication Channels:

- 📧 Email notifications
- 📲 SMS alerts for urgent updates
- 🔔 Mobile push notifications
- ✅ Delivery confirmation tracking

▼ 🎯 Capacity Management

Status	Action	Notification

▼ 📊 Admin Dashboard

Real-time metrics available:

🟢 Spots Available	Immediate confirmation	Confirmation email
🟡 Waitlisted	Queue position assigned	Waitlist notification
🔵 Spot Opens	Auto-promote from waitlist	Upgrade notification
🔴 Event Full	Close registration	Full capacity alert

The system automatically manages all capacity transitions without manual intervention.

- 👥 Total registrations count
- ⏱️ Waitlist queue length
- ✓ Check-in completion rate
- 📈 Historical attendance trends
- 🚫 Cancellation statistics

Quick Actions:

- Broadcast urgent updates to all participants
- Export registration data for records
- Generate QR codes for check-in

🔄 Complete System Architecture

```
graph TB
    subgraph "Attendee Interface"
    A["Registration Portal"]
    B["Personal Dashboard"]
    C["Mobile App"]
    end

    subgraph "Core System"
    D["Registration Engine"]
    E["Validation Module"]
    F["Notification Service"]
    G["Database"]
```

```
    end

    subgraph "Admin Interface"
    H["Organizer Dashboard"]
    I["QR Check-in System"]
    J["Analytics & Reports"]
    end

    A --> D
    B --> D
    C --> D
    D --> E
    E --> G
    D --> F
    F --> C
    F --> A
    H --> D
    I --> G
    J --> G
    G --> B
```

✨ **Benefits of This Interactive System:** Eliminates manual work, ensures data accuracy, provides instant communication, and creates a seamless experience for both attendees and organizers.

- [ ] Review system architecture with event organizers
- [ ] Identify technology stack for implementation
- [ ] Create detailed technical specifications
- [ ] Develop prototype for user testing